

# Aswin M Kumar

Thiruvananthapuram, Kerala, India

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## SUMMARY

Electrical and Electronics Engineering student with hands-on experience in embedded systems, IoT prototyping, power systems, and structural simulation. Built and deployed sensor-integrated hardware systems, completed government-funded aerospace training, and earned industry recognition for applied battery fault-detection research. Targeting embedded systems and IoT engineering roles.

## EDUCATION

### College of Engineering Trivandrum (CET)

Kerala, India

*B.Tech – Electrical and Electronics Engineering | CGPA: 8.21/10*

*2023 – Present (Graduating May 2027)*

- Relevant Coursework: Circuit Design, Embedded Systems, Control Systems, Power Systems, Signal Processing, IoT Applications

## TECHNICAL SKILLS

**Programming:** Python, C

**Hardware & Tools:** ESP32, Arduino, KiCAD, SolidWorks, MATLAB/Simulink, ANSYS Mechanical, ANSYS Fluent, Arduino Cloud, Wokwi, Git

**Domains:** Embedded Systems, IoT, PCB Design, Power Systems, Control Systems, Structural & CFD Simulation

**Protocols & Interfaces:** I2C, UART, GPIO, PWM

## EXPERIENCE

### IoT Intern

June 2025

*Keltron (Kerala State Electronics Development Corporation)*

*Thiruvananthapuram, Kerala*

- Prototyped IoT systems using ESP32 and Arduino Uno interfaced with ultrasonic, thermal, and LiDAR sensors
- Deployed cloud-connected monitoring systems on Arduino Cloud; implemented relay-based automation and real-time device state tracking
- Built functional prototypes – water level monitor, temperature sensor array, traffic signal controller – and authored technical report on IoT device architecture

### Participant – Drone/UAS Technologies Bootcamp

June 2025

*MeitY / SwaYam Programme, NIT Calicut*

*Calicut, Kerala*

- Completed 40-hour government-funded programme covering aerodynamics, drone assembly, flight operations, and computational fluid dynamics
- Executed ANSYS Mechanical and Fluent structural and CFD analyses on airfoil geometries; produced full simulation report
- Designed, fabricated, and flew a straight-wing fixed-wing glider; secured **3rd place** among 28+ teams in endurance flight competition

### Technical Team Member – Suspension & Steering

2023 – Present

*EVolve Racing CET (Formula Student / FSAE)*

*Thiruvananthapuram, Kerala*

- Supported suspension and steering system modelling using SolidWorks; contributed to cross-functional vehicle performance validation

### Membership Coordinator & Event Organiser

2025 – Present

*SAEINDIA – College of Engineering Trivandrum*

*Thiruvananthapuram, Kerala*

- Organised CADClash (10+ participants, first inter-college edition with AIT Pune) and Biomimicry ideathon at Tech Rave, SAEINDIA CET Technical Fest
- Managed SAEINDIA memberships and compliance documentation for BAJA SAEINDIA 2026

## PROJECTS

### LithiumSense – BMS Architecture | *System Design, Fault Analysis, Technical Presentation*

- Architected a fire-hardened Battery Management System with a 5-stage emergency response pipeline targeting sub-500ms fault response across overvoltage, undervoltage, overcurrent, short-circuit, and thermal runaway conditions
- Designed a 900°C-rated survivable black-box data module for post-incident forensic analysis of battery fault events
- Trophy winner** at Allianz Risk Consulting Compass-360 Expo; presented to industry risk and engineering professionals

### Off-Grid Solar Power Plant Design | *MATLAB, Simulink, PVGIS, MS Excel, LaTeX*

- Conducted on-site load survey for Machines Lab Block; estimated 93.17 kW connected load and 2,235 kWh daily energy demand under worst-case conditions
- Designed 545 kWp off-grid solar PV system with 5.2 MWh LiFePO4 battery storage (2-day backup), PV array of 1,010 modules, and 150 kW N+1 redundant inverter achieving performance ratio of 0.79
- Applied PVGIS irradiance data and battery DoD constraints for system sizing; completed techno-economic analysis (Rs. 9.94 Cr cost, ~15-year payback)

### Gesture-Controlled Interface | *Arduino Leonardo, MPU6050, I2C, C*

[GitHub](#)

- Implemented IMU-based motion control interface using MPU6050 gyroscope/accelerometer achieving ~50ms end-to-end latency for real-time gesture input

**Voice-Controlled Smart Switch** | *ESP32, Arduino IDE, Arduino Cloud, C*

- Built cloud-connected smart switch using ESP32 with speech-driven command processing and real-time device state monitoring via IoT cloud platform

**Trash Nada – AI Waste Management** | *Python, React, OpenCV, TensorFlow, IoT*

[GitHub](#)

- Built frontend and IoT integration layer for an AI-based waste classification system with real-time hotspot mapping
- **Top 10 / 133 teams** at Satwa25 National Hackathon (36-hour competition, 2025)

**NOT\_KTU Results** | *HTML5, CSS3, JavaScript, Tailwind CSS, Tone.js*

[GitHub](#)

- Satirical university portal parody with progressive password game and real-time audio feedback; **selected for TinkerHub Useless Projects 2.0**

**Resume CI/CD Pipeline** | *GitHub Actions, LaTeX, Git*

- Automated LaTeX resume compilation and PDF deployment on every commit via GitHub Actions; version-controlled continuous deployment

## ACHIEVEMENTS

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**Top 10 / 133 teams** – Satwa25 National Hackathon, Saintgits College of Engineering (2025)

**Trophy Winner** – Allianz Risk Consulting Compass-360 Expo, LithiumSense BMS Architecture (2025)

**3rd Place** – Endurance flight competition, MeitY/SwaYaan Drone/UAS Bootcamp, NIT Calicut (2025)

## CERTIFICATIONS

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Industrial Automation & PLC Systems – Bosch | Google AI Essentials – Google | Cybersecurity Essentials – Cisco | PCB Design Workshop – IET CET